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Vaporizing the Vacuum

HEAT NOTHINGNESS AND UNIFICATION

For as much as 95 percent of the universe's history, a cosmic correspondent concerned with the broad-brush, overall form of the universe would have reported more or less the same story: *Universe continues to expand. Matter continues to spread due to expansion. Density of universe continues to diminish. Temperature continues to drop. On largest of scales, universe maintains symmetric, homogeneous appearance.* But it wouldn't always have been so easy to cover the cosmos. The earliest stages would have required furiously hectic reporting, because in those initial moments the universe underwent rapid change. And we now know that what happened way back then has played a dominant role in what we experience today.

In this chapter, we will focus on critical moments in the first fraction of a second after the big bang, when the amount of symmetry embodied by the universe is believed to have changed abruptly, with each change launching a profoundly different epoch in cosmic history. While the correspondent can now leisurely fax in the same few lines every few billion years, in those early moments of briskly changing symmetry the job would have been considerably more challenging, because the basic structure of matter and the forces responsible for its behavior would have been completely unfamiliar. The reason is tied up with an interplay between *heat* and *symmetry,* and requires a complete rethinking of what we mean by the notions of empty space and of nothingness. As we will see, such rethinking not only enriches substantially our understanding of the universe's first moments, but also takes us a step closer to realizing a dream that harks back to Newton, Maxwell, and, in particular, Einstein—the dream of *unification.* Of equal importance, these developments set the stage for the most modern cosmological framework, *inflationary cosmology,* an approach that announces answers to some of the most